SYNTHETIC worked example - not biological validation

AlgaTwin 0.7.0 | 2026-10-25T17:49:34.873191+00:00

What this result means

The fitted curves describe these uploaded measurements. Independent biological validation is absent.

Intervals are conditional on your fixed parameters, initial conditions and specified measurement noise.

The selected parameters passed the numerical profile gate within your search bounds.

SYNTHETIC teaching data: numerical behavior only, not laboratory evidence.

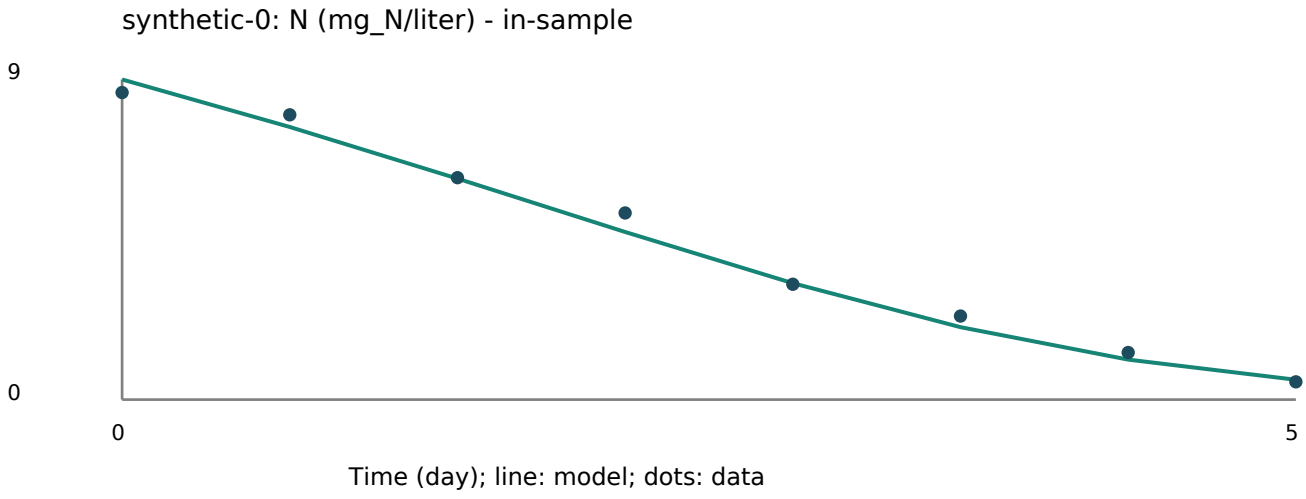
Parameter evidence

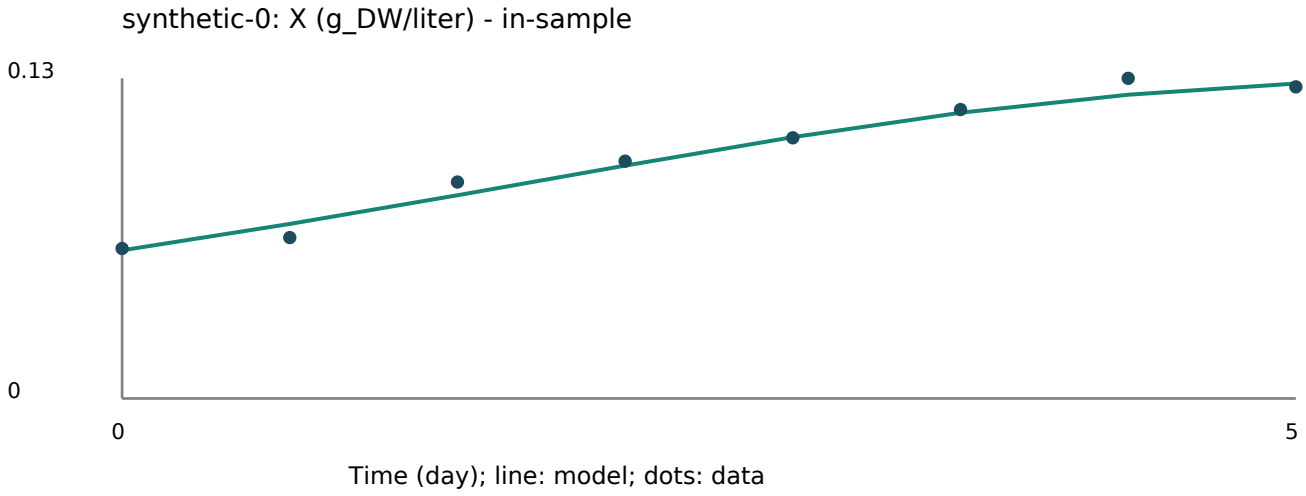
parameter	value	interval	unit	status
mu_max_day	0.901632	(0.899595, 0.90368)	1/day	IDENTIFIABLE

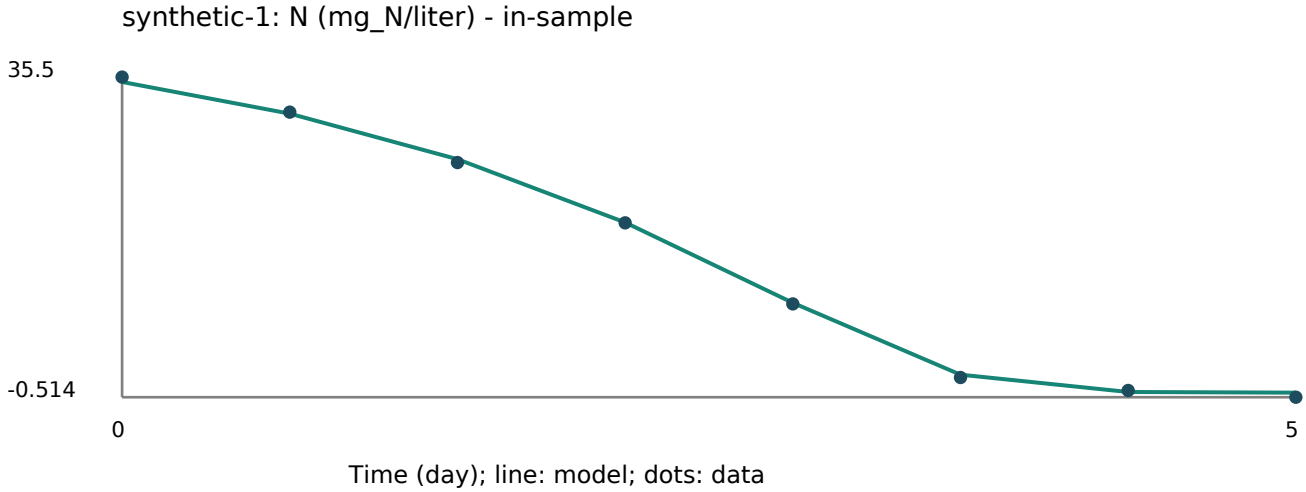
Confidence: 0.95. Method: Reoptimized profile likelihood; marginal chi-square(df=1) asymptotic cutoff; fixed noise scales

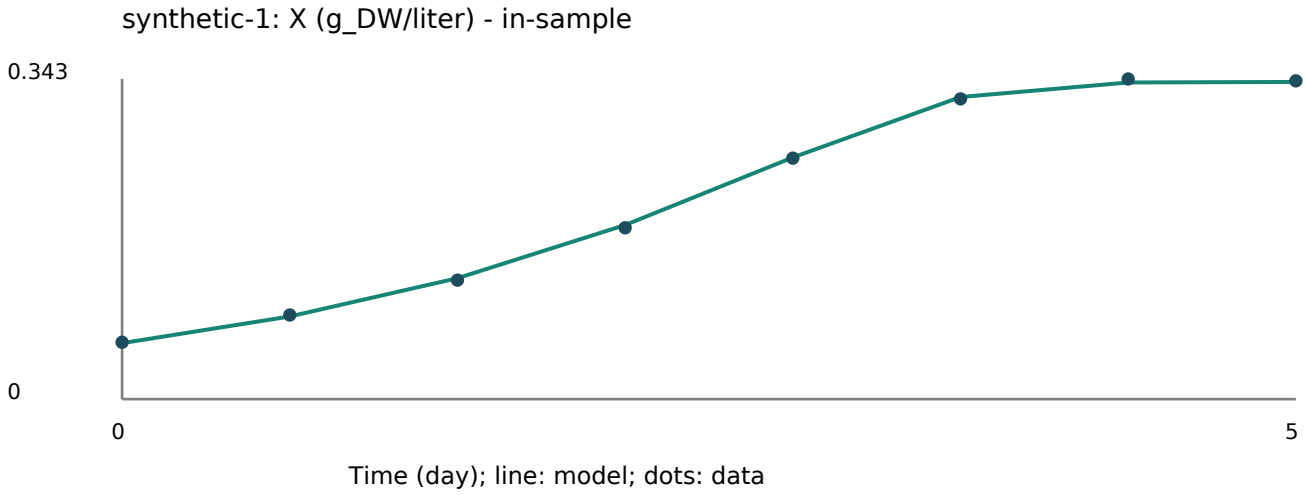
mu_max_day: Connected two-sided profile within supplied bounds, conditional on specified assumptions

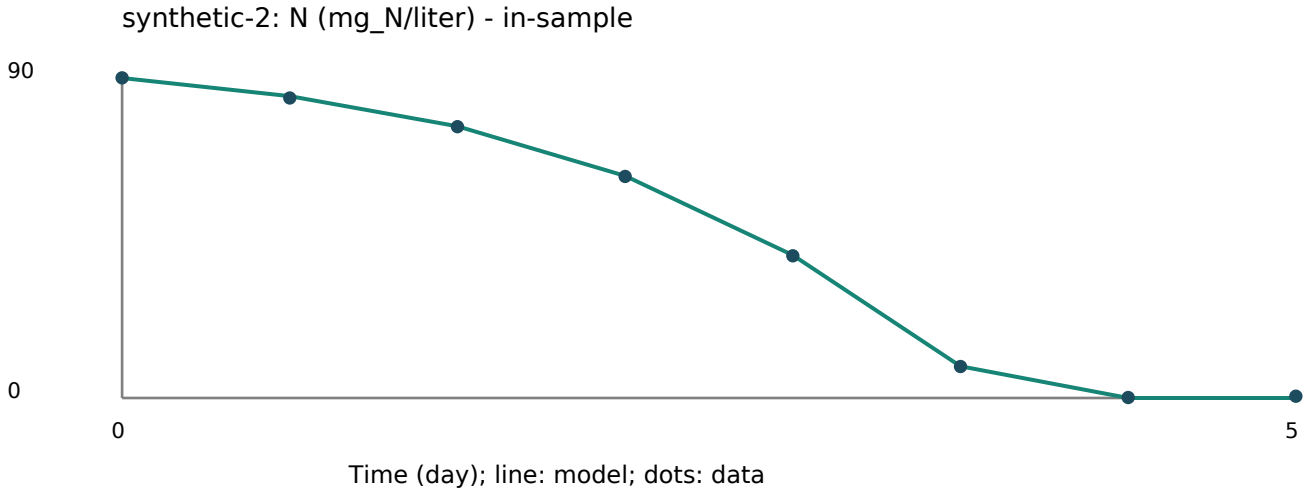
In-sample fit

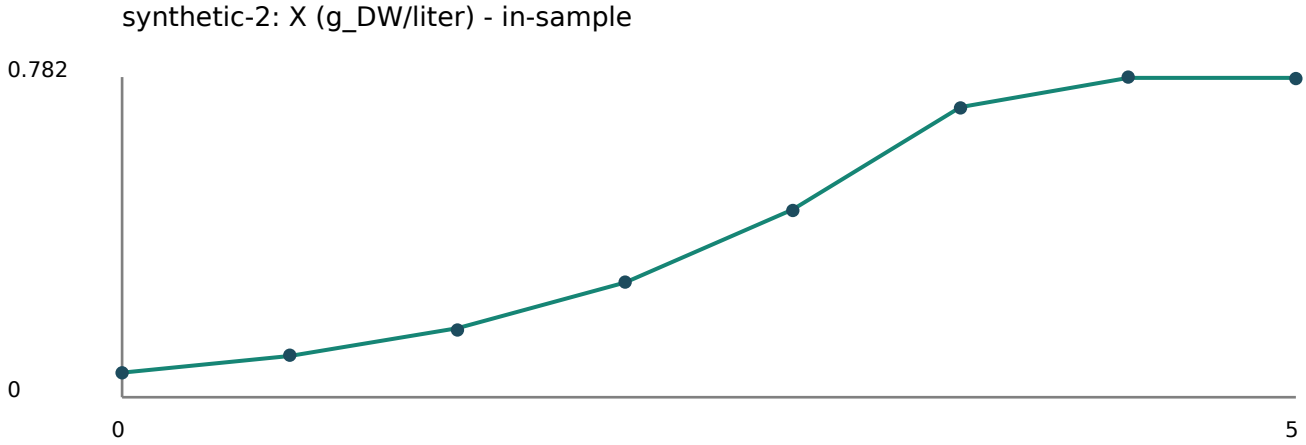





synthetic-1: N (mg_N/liter) - in-sample
35.5
-0.514
0
5
Time (day); line: model; dots: data


synthetic-1: X (g_DW/liter) - in-sample
0.343
0
0
5
Time (day); line: model; dots: data


synthetic-2: N (mg_N/liter) - in-sample
90
0
0
5
Time (day); line: model; dots: data



run_id	observable	split	unit	rmse	mae
synthetic-0	N	in-sample	mg_N/liter	0.291535	0.233748
synthetic-0	X	in-sample	g_DW/liter	0.00371586	0.00286757
synthetic-1	N	in-sample	mg_N/liter	0.332635	0.28372
synthetic-1	X	in-sample	g_DW/liter	0.00214094	0.00190475
synthetic-2	N	in-sample	mg_N/liter	0.277097	0.215505
synthetic-2	X	in-sample	g_DW/liter	0.00225103	0.00195718

Held-out evaluation

Leave-one-whole-run-out; refit and reprofile training runs only. All folds scored: True.

held_out	status	reason
synthetic-0	SCORED	Training profiles supported all fitted parameters
synthetic-1	SCORED	Training profiles supported all fitted parameters
synthetic-2	SCORED	Training profiles supported all fitted parameters

held_out_fold	run_id	observable	split	unit	rmse	mae
synthetic-0	synthetic-0	N	held-out	mg_N/liter	0.291583	0.233776
synthetic-0	synthetic-0	X	held-out	g_DW/liter	0.00371559	0.00286741
synthetic-0	synthetic-1	N	in-sample	mg_N/liter	0.332496	0.28347
synthetic-0	synthetic-1	X	in-sample	g_DW/liter	0.00214256	0.00190675
synthetic-0	synthetic-2	N	in-sample	mg_N/liter	0.276671	0.214599
synthetic-0	synthetic-2	X	in-sample	g_DW/liter	0.0022567	0.0019619
synthetic-1	synthetic-1	N	held-out	mg_N/liter	0.333105	0.28456
synthetic-1	synthetic-1	X	held-out	g_DW/liter	0.00213553	0.00189804
synthetic-1	synthetic-0	N	in-sample	mg_N/liter	0.291377	0.233656
synthetic-1	synthetic-0	X	in-sample	g_DW/liter	0.00371676	0.00286811
synthetic-1	synthetic-2	N	in-sample	mg_N/liter	0.278575	0.218536
synthetic-1	synthetic-2	X	in-sample	g_DW/liter	0.0022324	0.00194136
synthetic-2	synthetic-2	N	held-out	mg_N/liter	0.267785	0.194181
synthetic-2	synthetic-2	X	held-out	g_DW/liter	0.0024311	0.00209285
synthetic-2	synthetic-0	N	in-sample	mg_N/liter	0.292899	0.234537

held_out_fold	run_id	observable	split	unit	rmse	mae
synthetic-2	synthetic-0	X	in-sample	g_DW/liter	0.00370818	0.00286297
synthetic-2	synthetic-1	N	in-sample	mg_N/liter	0.328804	0.276518
synthetic-2	synthetic-1	X	in-sample	g_DW/liter	0.00218891	0.00196231

Assumptions and reproducibility

Original input SHA256: 1c1a652dfb4d147ab0073dde64e3831fe9f599e4a4f05bd7ffde099682f5bf60

Error model: {'kind': 'additive', 'scales': {'X': (0.003, 'g_DW/liter'), 'N': (0.3, 'mg_N/liter')}}

Search bounds (not literature ranges): {'mu_max_day': {'low': 0.3, 'high': 1.8, 'initial': 0.7, 'scale': 'log'}}

Run synthetic-0: SYNTHETIC

{"run_id": "synthetic-0", "origin": "SYNTHETIC", "input_sha256": "b3da7f7f71022553cb113ae50bdc88d88d24ad1897f35a6ea152f62c672bc988", "kind": "monod", "light": "monod", "temperature_c": null, "diurnal": false, "conditions": {"biomass_initial_g_L": 0.06, "nitrogen_initial_mg_L": 9, "incident_light_umol_m2_s": 60, "optical_path_m": 0.05}, "states": ["X", "N"], "state_units": ["g_DW/liter", "mg_N/liter"], "initial_state": [0.06, 9.0]}

parameter	value	unit	basis	status	tier
k_n_mg_L	5	mg_N/liter	elemental_nitrogen	UNVERIFIED	DEMO

species: SYNTHETIC; conditions: Teaching fixture, not measured species physiology; source: None; evidence: synthetic; locator: ; low: None; high: None

parameter	value	unit	basis	status	tier
extinction_a0	12	liter ** 1.0 / g_DW ** 1.0 / meter	dry_weight	UNVERIFIED	DEMO

species: SYNTHETIC; conditions: Teaching fixture, not measured species physiology; source: None; evidence: synthetic; locator: ; low: None; high: None

parameter	value	unit	basis	status	tier
background_extinction_m_inv	0.3	1/meter	medium	UNVERIFIED	DEMO

species: SYNTHETIC; conditions: Teaching fixture, not measured species physiology; source: None; evidence: synthetic; locator: ; low: None; high: None

parameter	value	unit	basis	status	tier
extinction_exponent	1	dimensionless	dimensionless	UNVERIFIED	DEMO

species: SYNTHETIC; conditions: Teaching fixture, not measured species physiology; source: None; evidence: synthetic; locator: ; low: None; high: None

parameter	value	unit	basis	status	tier
loss_rate_day	0	1/day	specific_growth	UNVERIFIED	DEMO

species: SYNTHETIC; conditions: Teaching fixture, not measured species physiology; source: None; evidence: synthetic; locator: ; low: None; high: None

parameter	value	unit	basis	status	tier
biomass_yield_g_per_gN	8	g_DW/g_N	dry_weight_per_nitrogen	UNVERIFIED	DEMO

species: SYNTHETIC; conditions: Teaching fixture, not measured species physiology; source: None; evidence: synthetic; locator: ; low: None; high: None

parameter	value	unit	basis	status	tier
mu_max_day	0.9	1/day	specific_growth	UNVERIFIED	DEMO

species: SYNTHETIC; conditions: Teaching fixture, not measured species physiology; source: None; evidence: synthetic; locator: ; low: None; high: None

parameter	value	unit	basis	status	tier
k_i_umol_m2_s	85	micromole/meter**2/second	photons	UNVERIFIED	DEMO

species: SYNTHETIC; conditions: Teaching fixture, not measured species physiology; source: None; evidence: synthetic; locator: ; low: None; high: None

Run synthetic-1: SYNTHETIC

{"run_id": "synthetic-1", "origin": "SYNTHETIC", "input_sha256": "34cdeb714a5988777aaa848d73d029d03fffa8c7572001252772bb853b478550", "kind": "monod", "light": "monod", "temperature_c": null, "diurnal": false, "conditions": {"biomass_initial_g_L": 0.06, "nitrogen_initial_mg_L": 35, "incident_light_umol_m2_s": 200, "optical_path_m": 0.05}, "states": ["X", "N"], "state_units": ["g_DW/liter", "mg_N/liter"], "initial_state": [0.06, 35.0]}

parameter	value	unit	basis	status	tier
k_n_mg_L	5	mg_N/liter	elemental_nitrogen	UNVERIFIED	DEMO

species: SYNTHETIC; conditions: Teaching fixture, not measured species physiology; source: None; evidence: synthetic; locator: ; low: None; high: None

parameter	value	unit	basis	status	tier
extinction_a0	12	liter ** 1.0 / g_DW ** 1.0 / meter	dry_weight	UNVERIFIED	DEMO

species: SYNTHETIC; conditions: Teaching fixture, not measured species physiology; source: None; evidence: synthetic; locator: ; low: None; high: None

parameter	value	unit	basis	status	tier
background_extinction_m_inv	0.3	1/meter	medium	UNVERIFIED	DEMO

species: SYNTHETIC; conditions: Teaching fixture, not measured species physiology; source: None; evidence: synthetic; locator: ; low: None; high: None

parameter	value	unit	basis	status	tier
extinction_exponent	1	dimensionless	dimensionless	UNVERIFIED	DEMO

species: SYNTHETIC; conditions: Teaching fixture, not measured species physiology; source: None; evidence: synthetic; locator: ; low: None; high: None

parameter	value	unit	basis	status	tier
loss_rate_day	0	1/day	specific_growth	UNVERIFIED	DEMO

species: SYNTHETIC; conditions: Teaching fixture, not measured species physiology; source: None; evidence: synthetic; locator: ; low: None; high: None

parameter	value	unit	basis	status	tier
biomass_yield_g_per_gN	8	g_DW/g_N	dry_weight_per_nitrogen	UNVERIFIED	DEMO

species: SYNTHETIC; conditions: Teaching fixture, not measured species physiology; source: None; evidence: synthetic; locator: ; low: None; high: None

parameter	value	unit	basis	status	tier
mu_max_day	0.9	1/day	specific_growth	UNVERIFIED	DEMO

species: SYNTHETIC; conditions: Teaching fixture, not measured species physiology; source: None; evidence: synthetic; locator: ; low: None; high: None

parameter	value	unit	basis	status	tier
k_i_umol_m2_s	85	micromole/meter**2 /second	photons	UNVERIFIED	DEMO

species: SYNTHETIC; conditions: Teaching fixture, not measured species physiology; source: None; evidence: synthetic; locator: ; low: None; high: None

Run synthetic-2: SYNTHETIC

{"run_id": "synthetic-2", "origin": "SYNTHETIC", "input_sha256": "0bb1f95c5dfde081b6d68f827168a340ed43fbecf964d5fe80bc4e3126b45a3c", "kind": "monod", "light": "monod", "temperature_c": null, "diurnal": false, "conditions": {"biomass_initial_g_L": 0.06, "nitrogen_initial_mg_L": 90, "incident_light_umol_m2_s": 500, "optical_path_m": 0.05}, "states": ["X", "N"], "state_units": ["g_DW/liter", "mg_N/liter"], "initial_state": [0.06, 90.0]}

parameter	value	unit	basis	status	tier
k_n_mg_L	5	mg_N/liter	elemental_nitrogen	UNVERIFIED	DEMO

species: SYNTHETIC; conditions: Teaching fixture, not measured species physiology; source: None; evidence: synthetic; locator: ; low: None; high: None

parameter	value	unit	basis	status	tier
extinction_a0	12	liter ** 1.0 / g_DW ** 1.0 / meter	dry_weight	UNVERIFIED	DEMO

species: SYNTHETIC; conditions: Teaching fixture, not measured species physiology; source: None; evidence: synthetic; locator: ; low: None; high: None

parameter	value	unit	basis	status	tier
background_extinction_m_inv	0.3	1/meter	medium	UNVERIFIED	DEMO

species: SYNTHETIC; conditions: Teaching fixture, not measured species physiology; source: None; evidence: synthetic; locator: ; low: None; high: None

parameter	value	unit	basis	status	tier
extinction_exponent	1	dimensionless	dimensionless	UNVERIFIED	DEMO

species: SYNTHETIC; conditions: Teaching fixture, not measured species physiology; source: None; evidence: synthetic; locator: ; low: None; high: None

parameter	value	unit	basis	status	tier
loss_rate_day	0	1/day	specific_growth	UNVERIFIED	DEMO

species: SYNTHETIC; conditions: Teaching fixture, not measured species physiology; source: None; evidence: synthetic; locator: ; low: None; high: None

parameter	value	unit	basis	status	tier
biomass_yield_g_per_gN	8	g_DW/g_N	dry_weight_per_nitrogen	UNVERIFIED	DEMO

species: SYNTHETIC; conditions: Teaching fixture, not measured species physiology; source: None; evidence: synthetic; locator: ; low: None; high: None

parameter	value	unit	basis	status	tier
mu_max_day	0.9	1/day	specific_growth	UNVERIFIED	DEMO

species: SYNTHETIC; conditions: Teaching fixture, not measured species physiology; source: None; evidence: synthetic; locator: ; low: None; high: None

parameter	value	unit	basis	status	tier
k_i_umol_m2_s	85	micromole/meter**2 /second	photons	UNVERIFIED	DEMO

species: SYNTHETIC; conditions: Teaching fixture, not measured species physiology; source: None; evidence: synthetic; locator: ; low: None; high: None